Canto · all results

Displaced-track reconstruction in HNL events (ACTS generic detector) and track reconstruction on TrackML · validation, one-time test, external sample, second population, training

	Canto					acorn, val-best builder			acorn, connected components (CC)		DeZoort et al.		published, HNL	published, TrackML			
	Canto-L 3 seeds · ttbar learned merging	Canto-H 3 seeds · ttbar walkthrough + merging	Canto-B 3 seeds · ttbar learned merging	Canto-L (HNL) 1 seed · HNL validation only learned merging τ 0.2	Canto-L (ttbar+HNL) 1 seed · ttbar+HNL validation only learned merging τ 0.3	acorn (HNL) 3 seeds · HNL learned merging	acorn (ttbar+HNL) 3 seeds · ttbar+HNL learned merging	acorn 3 seeds · ttbar learned merging	acorn (HNL) + CC 3 seeds · HNL prereg. cut	acorn + CC 3 seeds · ttbar prereg. cut = best cut	DeZoort IN 1 seed · ttbar lowest fake rate	DeZoort IN-L 1 seed · ttbar lowest fake rate	Wang et al. ACAT 2021 · HNL-trained attention GNN + DBSCAN	EggNet Calafiura et al. 2024 Chan et al. 2025	EC Calafiura et al. 2024	BC-HGNN Liu et al. 2026	EC-GNN Liu et al. 2026
HNL samples · validation pooled HNL val (hnl10_100, hnl15_200; 1,999 events) and ttbar val · mean over the seeds in the header · every model at a track fake rate ≤ FR* = 2.94 % where it can reach it · val values of selected configurations are measured on the events they were selected on																	
Displaced tracks |d0| ≥ 10 mm or r_v ≥ 100 mm · 1,331 per seed	99.10% ± 0.27	99.12% ± 0.25	98.45% ± 0.32	99.17% ± 0.32	**99.25%** ± 0.31	98.65% ± 0.31	97.90% ± 0.40	97.17% ± 0.44	93.84% ± 0.60	92.31% ± 0.97	84.82% ± 1.00	87.98% ± 0.94	—	—	—	—	—
Far-displaced tracks |d0| 100–300 mm · 167 per seed	97.60% ± 1.28	96.61% ± 1.32	96.21% ± 1.62	**98.80%** 165/167	**98.80%** 165/167	97.21% ± 1.32	95.61% ± 1.70	93.81% ± 2.22	93.41% ± 1.87	89.22% ± 3.42	80.24% ± 3.25	82.63% ± 3.02	—	—	—	—	—
Very far-displaced |d0| ≥ 300 mm · k/n · 19 per seed	87.7% 50/57	82.5% 47/57	78.9% 45/57	**94.7%** 18/19	**94.7%** 18/19	91.2% 52/57	84.2% 48/57	77.2% 44/57	80.7% 46/57	64.9% 37/57	47.4% 9/19	52.6% 10/19	—	—	—	—	—
Prompt muons W → μ in HNL events	**99.79%** ± 0.15	**99.79%** ± 0.15	99.71% ± 0.15	**99.79%** 1865/1869	**99.79%** 1865/1869	99.75% ± 0.16	99.75% ± 0.15	**99.79%** ± 0.15	98.15% ± 0.44	98.70% ± 0.35	91.17% ± 0.89	96.63% ± 0.58	—	—	—	—	—
Dense prompt tracks ttbar val, |d0| < 1 mm	96.31% ± 0.25	96.15% ± 0.16	96.57% ± 0.13	**96.69%** 50957/52702	96.41% 50809/52702	94.91% ± 0.16	95.18% ± 0.35	95.49% ± 0.24	82.10% ± 0.32	87.23% ± 0.59	62.63% ± 0.40	78.17% ± 0.31	—	—	—	—	—
Fake-track rate HNL events · lower is better	1.03% ± 0.05	1.43% ± 0.06	1.30% ± 0.05	1.02% 1 seed, no σ	**1.01%** 1 seed, no σ	1.37% ± 0.06	1.39% ± 0.07	1.53% ± 0.06	2.93% ± 0.08	2.97% ± 0.08	12.51% ± 0.19	6.09% ± 0.12	—	—	—	—	—
HNL samples · one-time blind test pooled HNL test (hnl10_100_test, hnl15_200_test, hnl15_100_external; 6,999 events) and ttbar_test · frozen models and working points, evaluated once · DeZoort was not evaluated on test																	
Displaced tracks |d0| ≥ 10 mm or r_v ≥ 100 mm · 5,013 per seed	99.08% ± 0.15	**99.18%** ± 0.13	98.43% ± 0.26	—	—	97.98% ± 0.20	97.37% ± 0.31	96.28% ± 0.56	92.57% ± 0.47	90.38% ± 1.02	—	—	—	—	—	—	—
Far-displaced tracks |d0| 100–300 mm · 584 per seed	**98.57%** ± 0.39	98.40% ± 0.43	97.03% ± 0.88	—	—	96.75% ± 0.59	95.49% ± 1.01	91.84% ± 1.80	91.72% ± 1.12	84.87% ± 2.74	—	—	—	—	—	—	—
Very far-displaced |d0| ≥ 300 mm · k/n · 36 per seed	87.0% 94/108	91.7% 99/108	89.8% 97/108	—	—	**93.5%** 101/108	90.7% 98/108	88.9% 96/108	90.7% 98/108	79.6% 86/108	—	—	—	—	—	—	—
Prompt muons W → μ in HNL events	99.77% ± 0.06	99.76% ± 0.06	**99.78%** ± 0.06	—	—	99.69% ± 0.07	99.65% ± 0.07	99.68% ± 0.08	97.62% ± 0.32	98.44% ± 0.16	—	—	—	—	—	—	—
Dense prompt tracks ttbar_test, |d0| < 1 mm	96.32% ± 0.24	96.22% ± 0.16	**96.58%** ± 0.13	—	—	94.84% ± 0.20	95.17% ± 0.34	95.40% ± 0.22	81.67% ± 0.33	87.41% ± 0.55	—	—	—	—	—	—	—
Fake-track rate HNL events · lower is better	**1.02%** ± 0.03	1.42% ± 0.04	1.29% ± 0.05	—	—	1.36% ± 0.03	1.38% ± 0.05	1.52% ± 0.03	2.86% ± 0.05	2.95% ± 0.05	—	—	—	—	—	—	—
Displaced tracks hnl10_100_test only · 603 per seed	**99.23%** ± 0.33	—	—	—	—	98.45% ± 0.47	97.51% ± 0.75	97.24% ± 0.65	92.70% ± 1.13	91.43% ± 1.23	—	—	—	—	—	—	—
Displaced tracks hnl15_200_test only · 549 per seed	**98.48%** ± 0.47	—	—	—	—	97.02% ± 0.70	96.17% ± 0.90	94.54% ± 1.02	91.56% ± 1.06	87.55% ± 2.05	—	—	—	—	—	—	—
Wang et al.'s evaluation sample · hnl15_100_external m_N = 15 GeV, cτ = 100 mm, 4,999 events · test only, never used in training or selection · all-track, prompt-muon and HNL-daughter rows under Wang et al.'s definitions (Canto, the CC columns and Wang et al.'s published values)																	
Displaced tracks |d0| ≥ 10 mm or r_v ≥ 100 mm · 3,861 per seed	**99.14%** ± 0.17	—	—	—	—	98.04% ± 0.22	97.51% ± 0.28	96.37% ± 0.61	92.70% ± 0.56	90.62% ± 0.98	—	—	—	—	—	—	—
All tracks Wang et al.'s definition · 116,449 per seed	99.14% ± 0.05	**99.15%** ± 0.04	99.13% ± 0.04	—	—	—	—	—	94.26% ± 0.34	95.66% ± 0.14	—	—	95.61% ± 0.06	—	—	—	—
Prompt muons Wang et al.'s definition: W muons, ≥ 5 hits · 4,744 per seed	99.65% ± 0.09	99.63% ± 0.09	**99.66%** ± 0.09	—	—	—	—	—	97.46% ± 0.30	98.17% ± 0.18	—	—	99.0% ± 0.2	—	—	—	—
HNL daughters Wang et al.'s 'displaced', all |d0| · 5,623 per seed	99.22% ± 0.15	**99.35%** ± 0.11	98.77% ± 0.19	—	—	—	—	—	93.31% ± 0.57	92.12% ± 0.74	—	—	91.6% ± 0.5 their displaced selection	—	—	—	—
Fake-track rate lower is better	**1.03%** ± 0.03	1.44% ± 0.04	1.30% ± 0.05	—	—	—	—	—	2.89% ± 0.05	2.95% ± 0.05	—	—	not reported	—	—	—	—
Second population · not used in any selection K0S and Λ daughters (V0 daughters) in the HNL events, photon-conversion e± in ttbar · targets pT > 1 GeV, |η| < 4, ≥ 5 hits · every model frozen at its working point																	
V0 daughters HNL val (hnl10_100 + hnl15_200) · 2,197 per seed	**99.06%** ± 0.24	98.92% ± 0.26	99.04% ± 0.26	—	—	98.26% ± 0.40	98.47% ± 0.33	98.33% ± 0.34	89.47% ± 0.91	91.20% ± 1.02	—	—	—	—	—	—	—
V0 daughters HNL test (hnl10_100 + hnl15_200) · 1,925 per seed	98.49% ± 0.44	98.60% ± 0.43	**98.74%** ± 0.40	—	—	97.97% ± 0.45	98.03% ± 0.54	98.27% ± 0.42	88.35% ± 1.57	90.72% ± 1.09	—	—	—	—	—	—	—
Conversion e± ttbar val · 1,892 per seed	78.42% ± 1.25	72.74% ± 1.45	**78.88%** ± 1.38	—	—	73.40% ± 1.20	75.85% ± 1.27	76.66% ± 1.22	44.03% ± 2.86	51.76% ± 1.95	—	—	—	—	—	—	—
Conversion e± ttbar_test · 1,753 per seed	78.27% ± 1.30	72.94% ± 1.51	**78.61%** ± 1.36	—	—	74.25% ± 1.36	75.70% ± 1.46	76.29% ± 1.44	44.29% ± 2.62	51.40% ± 2.12	—	—	—	—	—	—	—
TrackML throughput phase · one-time test pileup 200 in every condition; 'no noise' removes only the noise hits, TrackML-1GeV keeps only the hits of particles with pT > 1 GeV (both using truth) · 300 test events · Canto-L retrained on 1,399 throughput-phase events, one seed, working point WP-A · Canto-definition rows: Canto-L only · then each reference's rows under its own evaluation code (ported; recounted after the freeze), published values on the authors' own events																	
Target efficiency all hits · Canto definition	**99.65%** ± 0.01	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Target efficiency no noise · Canto definition	**99.66%** ± 0.01	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Target efficiency TrackML-1GeV · Canto definition	**99.77%** ± 0.01	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Fake-track rate all hits · Canto definition lower is better	**1.68%** ± 0.01	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Fake-track rate no noise · Canto definition lower is better	**1.29%** ± 0.01	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Fake-track rate TrackML-1GeV · Canto definition lower is better	**0.055%** ± 0.004	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
TrackML score all hits · challenge weights	**0.966** ± 0.0001	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
TrackML score no noise · challenge weights	**0.968** ± 0.0001	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Track efficiency TrackML-1GeV · EggNet code	**99.56%** ± 0.01	—	—	—	—	—	—	—	—	—	—	—	—	**99.56%** ± 0.03	98.98% ± 0.09	—	—
Fake-track rate TrackML-1GeV · EggNet code lower is better	**0.008%** ± 0.001	—	—	—	—	—	—	—	—	—	—	—	—	0.06%	0.12%	—	—
Track efficiency TrackML-1GeV · Liu et al. code	**99.77%** ± 0.01	—	—	—	—	—	—	—	—	—	—	—	—	—	—	98.55%	—
Fake rate TrackML-1GeV · Liu et al. code lower is better	**0.26%** ± 0.01	—	—	—	—	—	—	—	—	—	—	—	—	—	—	1.23%	—
Track efficiency full TrackML · EggNet code	**99.41%** ± 0.01	—	—	—	—	—	—	—	—	—	—	—	—	95.87% ± 0.03	—	—	—
Fake-track rate full TrackML · EggNet code lower is better	2.40% ± 0.01	—	—	—	—	—	—	—	—	—	—	—	—	**0.19%** ± 0.01	—	—	—
Track efficiency full TrackML · Liu et al. code	**99.65%** ± 0.01	—	—	—	—	—	—	—	—	—	—	—	—	—	—	97.86%	96.35%
Fake rate full TrackML · Liu et al. code lower is better	**34.94%** ± 0.11	—	—	—	—	—	—	—	—	—	—	—	—	—	—	36.71%	55.58%
Training · edge classifier the CC columns use the networks of the acorn columns · last line of the Canto-L cells: the TrackML instance (retrained)																	
Training data events, one graph per event	ttbar 7,999 events TrackML: 1,399	ttbar 7,999 events	ttbar 7,999 events	HNL signal 15,999 events	ttbar + HNL 15,998 events	HNL signal 15,999 events	ttbar + HNL 15,998 events	ttbar 7,999 events	HNL signal 15,999 events	ttbar 7,999 events	ttbar 7,999 events	ttbar 7,999 events	—	—	—	—	—
Parameters edge classifier	347,777 TrackML: 347,777	347,777	347,777	347,777	347,777	263,937	263,937	263,937	263,937	263,937	6,448	127,126	—	—	—	—	—
Training steps one graph (TrackML: one window) per step	239,970 30 epochs TrackML: 120,000	239,970 30 epochs	239,970 30 epochs	479,970 30 epochs	479,940 30 epochs	479,970 30 epochs	479,940 30 epochs	239,970 30 epochs	479,970 30 epochs	239,970 30 epochs	60,000 15 epochs	60,000 15 epochs	—	—	—	—	—

Samples and definitions.
ACTS generic detector simulation without pileup: HNL signal samples (m_N, cτ in the sample name) and ttbar. Track efficiency by double-majority matching: more than half of the track's hits come from the particle and more than half of the particle's hits are on the track. Targets: pT > 1 GeV, |η| < 4, at least 5 hits. Displaced tracks: HNL-daughter targets with |d0| ≥ 10 mm or r_v ≥ 100 mm. Fake-track rate: the fraction of tracks with at least 5 hits that match no particle. TrackML: pileup 200, with the definitions named in each row.

Working points.
HNL samples: every model at a track fake rate ≤ FR* = 2.94 % on pooled HNL val (preregistered); a model or seed that cannot reach FR* is at its lowest fake rate. Canto: the builder of each branch chosen on val by rule R6 with the one-standard-error rule (within one binomial standard error of the best efficiency, the lowest fake rate); Canto-L selected by rule R8 and frozen before the test (results/final_selection_v1.json). acorn (HNL), acorn (ttbar+HNL) and acorn: acorn's interaction network with the val-best builder of the full menu, chosen per seed (MWS-layer + learned merging for every seed). acorn (HNL) + CC and acorn + CC: connected components at the preregistered cut rule, the reference of the preregistered claims; for acorn it equals the best cut ≤ FR*. DeZoort IN and IN-L: one seed at the screening budget; they cannot reach FR*. Canto-L (HNL) and Canto-L (ttbar+HNL): the Canto-L pipeline trained on HNL signal events or on ttbar + HNL, one seed each, with learned merging at the setting chosen on val by the builder rule at FR* (header); evaluated on validation only, not on test, the external sample, the second population or TrackML. TrackML: working point WP-A, chosen on 150 val events (the highest target efficiency at a track fake rate ≤ FR*, the HNL value, not re-tuned; one-standard-error tie rule).

Seeds and uncertainties.
HNL samples: means over the training seeds in the column header; k/n pooled over the seeds. Second line of a cell: the combined uncertainty σ = (σ_boot² + s²/n)^½ of the mean, in percentage points (score units in score rows), with σ_boot from an event bootstrap (B = 1,000 on val, 500 on test) and s the standard deviation over the n seeds. The |d0| ≥ 300 mm rows show k/n instead (statistics limited). Canto-L (HNL) and Canto-L (ttbar+HNL): one seed; the σ of their displaced-track row takes the seed standard deviation of Canto-L (trained on ttbar) as its seed term (results/training_sample_bins_v1.md); their other rows have no σ in the source and show k/n, or '1 seed, no σ' for the fake rate. TrackML: Canto-L has one training seed, σ is its event-bootstrap error (B = 500). Published values carry the uncertainty quoted by their authors; none is shown where none is quoted (Liu et al.; Calafiura et al.'s fake rates).

Shading.
Framed in orange: the first place of each row, the highest value (the lowest for rates; cells showing the same value share it), in whichever column it is; published values take part like any other column. A difference below 2σ, σ the two cells' uncertainties in quadrature, is not resolved. Paired differences, with smaller uncertainties, are in results/strong_baselines_v3_differences.csv and results/bootstrap_v1.csv.

External sample and Wang et al.
hnl15_100_external is Wang et al.'s evaluation sample (arXiv:2203.08800), 4,999 events; it was evaluated once and never used in training or selection. Column Wang et al.: their published values on this sample (ACAT 2021; attention GNN + DBSCAN trained on HNL events), as reported in the paper (docs/paper_facts.md). Definitions: all tracks, every target, whose count matches the one their binomial error implies; prompt muons, W muons with at least 5 hits, their prompt denominator. HNL daughters: all daughter targets for our models; Wang et al.'s displaced value comes from their own selection of these particles (about half of them, not reproducible from the paper), so their cell is shown as reported and is not a like-for-like comparison. Wang et al. report no fake rate. Of our models only Canto and the CC columns were evaluated under these definitions. acorn (HNL) + CC at the fixed cut of our earlier reproduction gives 0.9554 for all tracks, against their 0.9561.

Second population.
V0 daughters (charged daughters of K0S and Λ decays) exist only in the HNL samples; the external sample carries no parent labels and is not used for them. acorn (HNL) and acorn (ttbar+HNL) saw V0 daughters in training; Canto and acorn, trained on ttbar, did not. Conversion e±: targets from photon conversions in ttbar. No row of this band entered any selection.

TrackML throughput phase.
Canto-L is retrained on TrackML (1,399 throughput-phase training events, one seed, run canto_l_v1/seed0_x1); no other column was run on TrackML. Three conditions, all at pileup 200: all hits (the primary condition: training and every choice were made here), no noise (noise hits removed using truth) and TrackML-1GeV (only the hits of particles with pT > 1 GeV, using truth). The frozen pipeline was evaluated once on 300 test events. Canto-definition rows (Canto-L only): the HNL definitions above, targets pT > 1 GeV, |η| < 4, at least 5 hits, from the test reports; the TrackML score is not computed in TrackML-1GeV. Reference rows: recounted after the freeze from the frozen pipeline's test tracks with a port of the reference's evaluation code (EggNet; Liu et al.), checked against the unmodified code. Only references that used the throughput phase are compared. Liu et al.'s FR is a model-to-model comparison, not final track-building performance; Canto-L's is after its final builder. Not shown as rows, duplicate rates under the EggNet code: Canto-L 0.13 % (TrackML-1GeV) and 0.17 % (full TrackML); EggNet 1.29 % and 2.76 %; EC 4.21 % (TrackML-1GeV).

Training.
Edge classifier only (results/model_table_v1.md; TrackML: the run's training record); the metric-learning graph stage is not included. One step is one event graph on the HNL samples and one training window on TrackML. acorn (HNL), acorn (ttbar+HNL), Canto-L (HNL) and Canto-L (ttbar+HNL) train on events of the evaluated HNL samples (their training split); every other model on ttbar only.